ELM 361 Uygulama -3-

3.16 (SSB)

An SSB-AM signal is generated by modulating an 800 kHz carrier by the signal $m(t) = \cos(2000\pi t) + 2\sin(2000\pi t)$. The amplitude of the carrier is $A_c = 100$.

1. Determine the signal $\hat{m}(t)$.
2. Determine the (time-domain) expression for the lower sideband of the SSB-AM signal.
3. Determine the magnitude spectrum of the lower-sideband SSB signal.

---

1. $\hat{m}(t) = m_h(t)$ (hilbert transform)

$$\hat{m}(t) = \cos(2000\pi t - \pi/2) + 2\sin(2000\pi t - \pi/2)$$

$$\hat{m}(t) = \sin(2000\pi t) - 2\cos(2000\pi t)$$

2.

$$y_{LSB}(t) = A_c\left(m(t)\cdot\cos(2\pi f_c t) + \hat{m}(t)\cdot\sin(2\pi f_c t)\right)$$

$A_c = 100$, $f_c = 800\text{kHz}$, $m(t) = \cos(2000\pi t) + 2\sin(2000\pi t)$

$$y_{LSB}(t) = 100\cdot\left(\left(\cos(2000\pi t) + 2\sin(2000\pi t)\right)\cdot\cos(2\pi 800k\cdot t) + \left(\sin(2000\pi t) - 2\cos(2000\pi t)\right)\cdot\sin(2\pi 800k\cdot t)\right)$$

$$= 100\cdot\left(\cos(\underbrace{2\pi f_m}_{2000\pi t})\cdot\cos(\underbrace{2\pi f_c}_{2\pi 800k\, t}) + 2\sin(\underbrace{2\pi f_m}_{2000\pi t})\cdot\cos(\underbrace{2\pi f_c}_{2\pi 800k\, t}) + \sin(\underbrace{2\pi f_m}_{2000\pi t})\cdot\sin(\underbrace{2\pi f_c}_{2\pi 800k\, t}) - 2\cos(\underbrace{2\pi f_m}_{2000\pi t})\cdot\sin(\underbrace{2\pi f_c}_{2\pi 800k\, t})\right)$$

$f_c - f_m = f_1$, $f_c + f_m = f_2$

$$= 100\cdot\left(\frac{1}{2}\cdot\left(\cos(2\pi(f_m + f_c)t) + \cos(2\pi(f_m - f_c)t)\right) + \frac{1}{2}\cdot\left(\sin(2\pi(f_m + f_c)t) + \sin(2\pi(f_m - f_c)t)\right) + \frac{1}{2}\cdot\left(\cos(2\pi(f_m - f_c)t) - \cos(2\pi(f_m + f_c)t)\right) - \frac{1}{2}\cdot\left(\sin(2\pi(f_c + f_m)t) - \sin(2\pi(f_c - f_m)t)\right)\right)$$

$$= \sin(2\pi(f_m - f_c)t)$$

(1)

$$y_{LSB}(t) = 100.\left(\cos\left(2\pi(f_m - f_c)t\right) + 2\sin\left(2\pi(f_m - f_c)t\right)\right)$$

$$= 100.\left(\cos\left(2\pi(1000 - 800k)t\right) + 2\sin\left(2\pi(1000 - 800k)t\right)\right)$$

$$y_{LSB}(t) = 100\left(\cos\left(2\pi(\underbrace{1000 + 800k}_{f_c})t\right) - 2\sin\left(2\pi(\underbrace{800k - 1000}_{f_c})t\right)\right)$$

3.

$Y_{LSB}(f) = ?$

$$y_{LSB}(t) = 100.\left(\cos\left(2\pi(f_c - f_m)t\right) + 2\sin\left(2\pi(f_c - f_m)t\right)\right)$$

Hint: cos

$\cos\alpha.\cos\beta = \frac{1}{2}.(\cos(\alpha+\beta)+\cos(\alpha-\beta))$

$\cos\alpha.\sin\beta = \frac{1}{2}.(\sin(\beta+\alpha)+\sin(\beta-\alpha))$

$\sin\alpha.\cos\beta = \frac{1}{2}(\sin(\alpha-\beta)+\sin(\alpha+\beta))$

$\sin\alpha.\sin\beta = \frac{1}{2}.(\cos(\alpha-\beta)-\cos(\alpha+\beta))$

$$Y_{LSB}(f) = 100.\frac{1}{2}.\left(\delta(f-(f_c-f_m)) + \delta(f+(f_c-f_m))\right)$$

$$-\frac{100}{j}2\left(\delta(f-(f_c-f_m)) - \delta(f+(f_c-f_m))\right)$$

$$Y_{LSB}(f) = 50.\left(\delta(f-800k+1000) + \delta(f+800k-1000)\right)$$

$$+100j\left(\delta(f-800k+1000) - \delta(f+800k-1000)\right)$$

$$= (50+100j)\left(\delta(f-800k+1000)\right) + (50-100j).\left(\delta(f+800k-1000)\right)$$

↓ magnitude spectrum

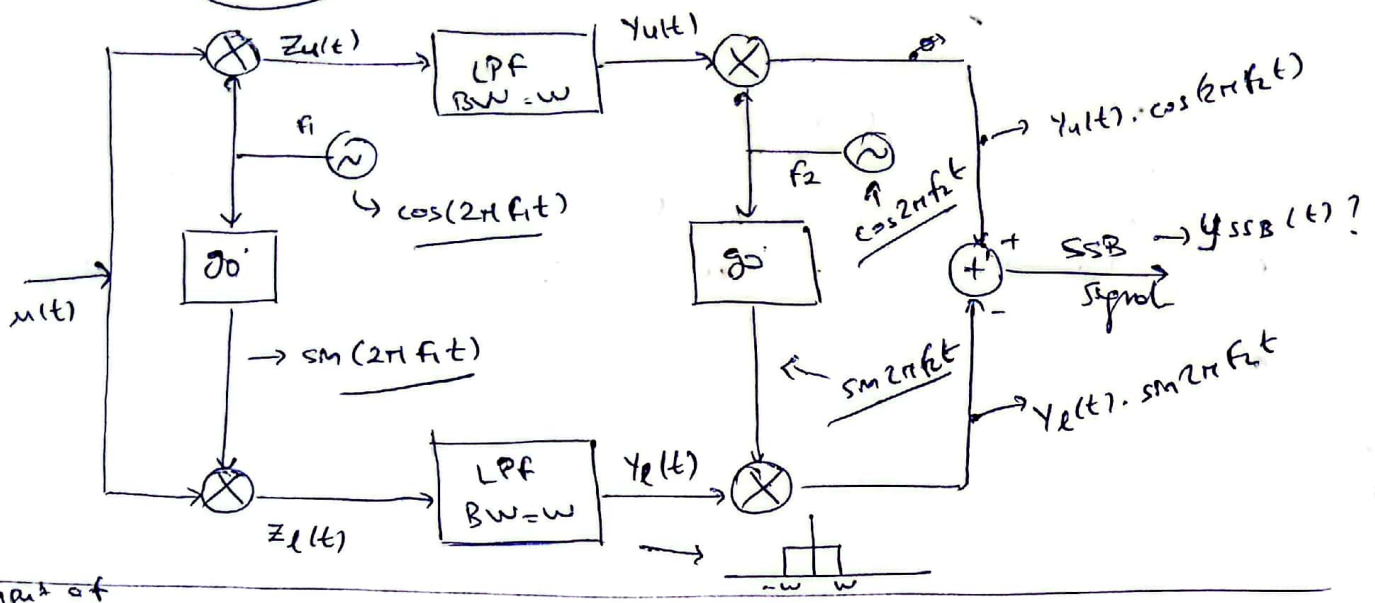
$$|Y_{LSB}(f)| = \sqrt{50^2+100^2}.\left(\delta(f-800k+1000) + \delta(f+800k-1000)\right)$$

$$= 10\sqrt{125}.\left(\delta(f-800k+1000) + \delta(f+800k-1000)\right)$$

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3.17

Weaver's SSB modulator is illustrated in figure P-3.17. By taking the input signal as $m(t) = \cos 2\pi f_m t$ where $f_m < W$ demonstrate that by proper choice of (f_1 and f_2) the output is an SSB signal



input of upper LPF

$$z_u(t) = m(t)\cdot\cos(2\pi f_1 t)$$

$$= \cos(2\pi f_m t)\cdot\cos(2\pi f_1 t) = \frac{1}{2}\cdot\left[\cos(2\pi(\underbrace{f_1 - f_m}_{f_a})t) + \cos(2\pi(\underbrace{f_1+f_m}_{f_b})t)\right]$$

input of lower LPF

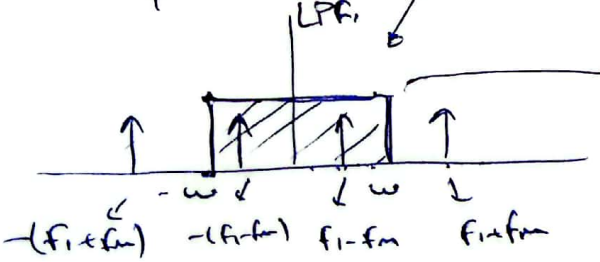
$$z_l(t) = \cos(2\pi f_m t)\cdot\sin(2\pi f_1 t)$$

$$= \frac{1}{2}\cdot\left[\sin(2\pi(f_1 - f_m)t) + \sin(2\pi(f_1+f_m)t)\right]$$

$f_m < W$ → we will select f_1.

↳ $|f_1 - f_m| < W$ and $f_1 + f_m > W$ → SSB obtained easily.

$$Z_u(f) = \frac{1}{4}\cdot\left(\delta(f-f_a) + \delta(f+f_a)\right) + \frac{1}{4}\cdot\left(\delta(f-f_b) + \delta(f+f_b)\right)$$



$$y_u(t) = \frac{1}{2}\cos(2\pi(f_1 - f_m)t)$$

$$y_l(t) = \frac{1}{2}\sin(2\pi(f_1 - f_m)t)$$

$$y_{SSB}(t) = \cancel{y_u(t) - y_l(t)} =$$

$$= y_u(t)\cdot\cos(2\pi f_2 t) - y_l(t)\cdot\sin(2\pi f_2 t)$$

$$= \frac{1}{2}\cdot\cos(2\pi(f_1 - f_m)t)\cdot\cos(2\pi f_2 t) - \frac{1}{2}\cdot\sin(2\pi(f_1 - f_m)t)\cdot\sin(2\pi f_2 t)$$

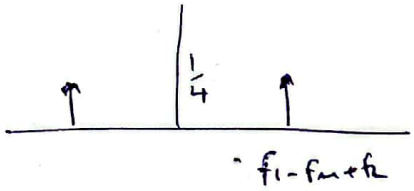
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$$y_{SSB}(t) = \frac{1}{4}\cdot\left(\cos(2\pi(f_1 - f_m + f_2)t) + \cos(2\pi(f_1 - f_m - f_2)t)\right)$$

$$-\frac{1}{4}\cdot\left(\cos(2\pi(f_1 - f_m - f_2)t) - \cos(2\pi(f_1 - f_m + f_2)t)\right)$$

$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$

$$y_{SSB}(t) = \frac{1}{2}\cdot\cos(2\pi(f_1 - f_m + f_2)t)$$



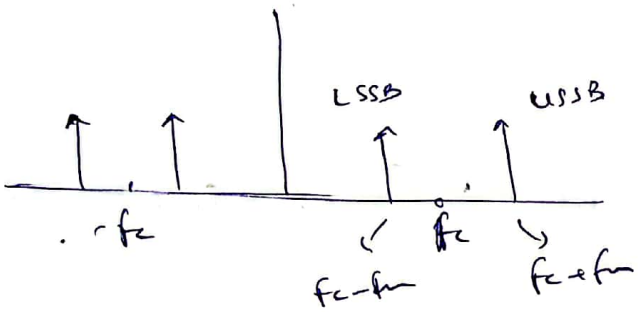
at f_c

$f_1 + f_2 - f_m = f_c + f_m \rightarrow$ USSB (Upper)

$f_1 + f_2 - f_m = f_c - f_m \rightarrow$ LSSB (lower)

$f_c = -(f_1 + f_2)$ $\quad f_c = (f_1 + f_2)$.

we are choosing different f_1 and f_2 for USSB and LSSB.



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3.24 (FDM)

We wish to transmit 60 voice-band signals by SSB (upper-sideband) modulation and freq. division multiplexing (FDM). Each of the 60 signals has a spectrum as shown figure P-3.24. Note that the voiceband signal is bandlimited to 3kHz. If each signal is frequency translated separately, we require a frequency synthesizer that produces 60 carrier frequencies to perform the FDM. On the other hand, if we subdivide the channels into L groups of K subchannels each, such that $LK = 60$, we may reduce the number of frequencies from the synthesizer to $L+K$.

1) Illustrate the spectrum of the SSB signals in a group of K subchannels. Assume that a 1kHz guardband separates the signals in adjacent frequency subchannels and that the carrier frequencies are $f_{c_1} = 10\text{kHz}$, $f_{c_2} = 14\text{kHz}$ – etc.

2) Sketch L and K such that $LK = 60$ and $L+K$ is a minimum.

3) Determine the frequencies of the carriers if the 60 FDM signals occupy the frequency band 300-540 kHz, and each group of K signals occupies the band 10kHz to $(10+4K)$ kHz.

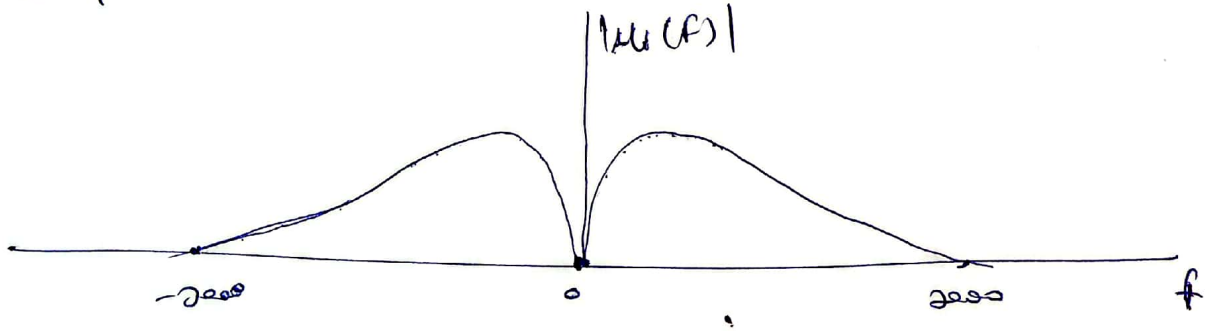
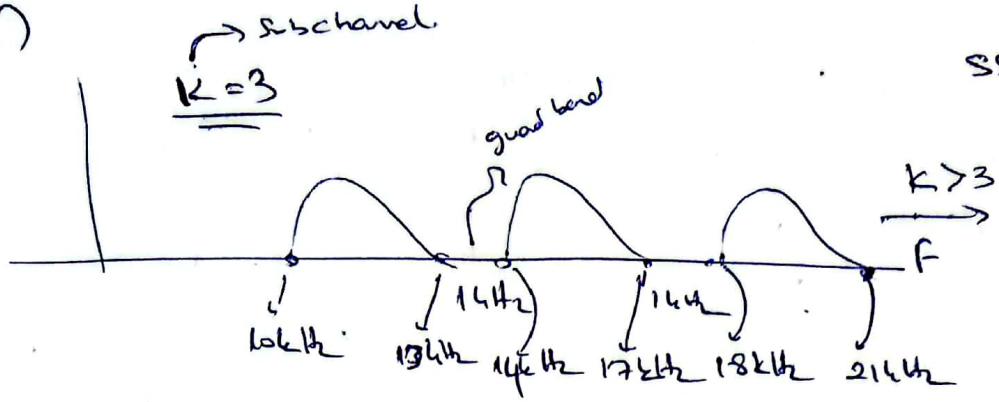


Figure 3.24

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1)



SSB → USSB

upper sideband

2) groups

K=60 (L, K) or (K, L) → (1,60), (2,30), (3,20)

↓ (4,15), (6,10).

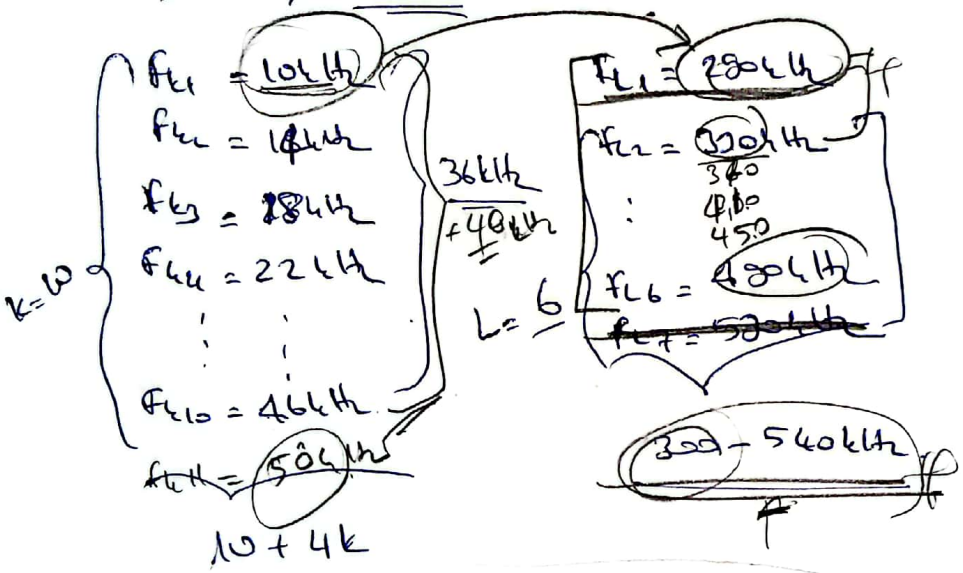
subchanels

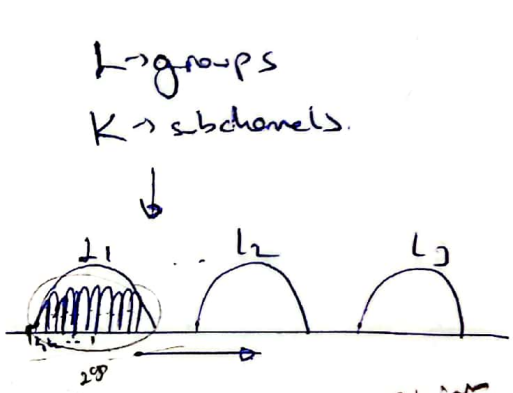
Minimum value → L+k

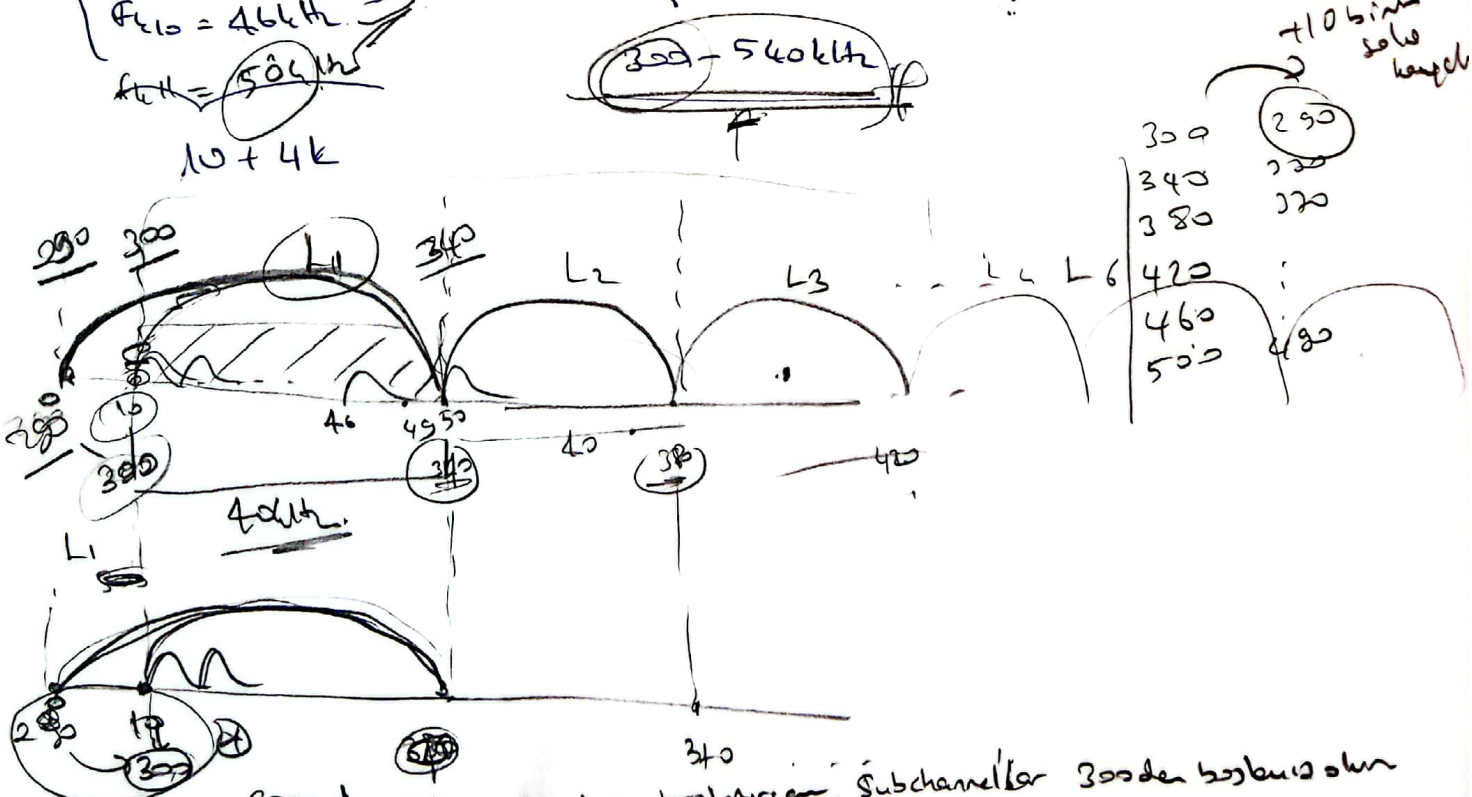
↓ ↓

6 + 10 → 16.

3) L=6, K=10 we need 16 carriers with frequencies.







Eğer L1 probunu 290'dan başlatırsam, subchannel'lar 300'den başlamış olur

4.9 (FM)

The carrier $c(t) = 100\cos(2\pi f_c t)$ is frequency modulated by the signal $m(t) = 5\cos(20000\pi t)$ where $f_c = 10^8$ Hz.

The peak frequency deviation is 20 kHz.

1.) Determine the amplitude and frequency of all signal components that have a power level of at least 10% of the power of the unmodulated carrier component.

2.) From carson's rule, determine the approximate BW of FM signal.

$$y(t) = A_c \sum_{n=-\infty}^{+\infty} J_n(\beta)\cdot\cos(2\pi(f_c + n\cdot f_m)t)$$

$A_c = 100$ $\quad f_m = 10000$ Hz $= 10^4$ Hz

$f_c = 10^8$ Hz.

$$\beta = \frac{\Delta f}{B} = \frac{\Delta f \cdot (m_p)}{f_m} = \frac{\Delta f_{max}}{f_m}$$

(m_p : $\max|m(t)|$)

$$f_m = \frac{20\cdot 10^3}{10^4} = 2$$

$\beta = 2$

$$y(t) = 100\sum_{n=-\infty}^{+\infty} J_n(2)\cdot\cos(2\pi(10^8 + n\cdot 10^4)t)$$

1) unmodulated carrier component → $c(t)$ → $P_c = \frac{A_c^2}{2} = \frac{100^2}{2} = 5000$

at least 10% → $5000\cdot\frac{10}{100} = 500$ → $P_{J(n)} \geq 500$

$$P_{J(n)} = \frac{(A_c J_n(2))^2}{2}$$

Index n	$J_n(2)$	freq (Hz)	Amplitude $100\,J_n(2)$	Power →
0	.2239	10^8	22.39	250.63
1	.5762	$10^8 \pm 10^4$	57.69	1663.1
2	.3528	$10^8 \pm 2\cdot10^4$	35.28	622.46
3	.1289	$10^8 \pm 3\cdot10^4$	12.89	83.05
4	.0340	$10^8 \pm 4\cdot10^4$	3.40	5.7785

index → n = 0, 1

Hz given → × ×100

index	freq (Hz)	Amp Mode
0	$10^8 + 10^4$	57.67
1	$10^8 + 2.10^4$	35.28
-1	$10^8 - 10^4$	57.67
-2	$10^8 - 2.10^4$	35.28

$$J_n^2(2) = J_{-n}^2(2)$$

2) $B_c = 2B(\beta+1) = 2.10^4(2+1) = 6.10^4 \text{ Hz}$

↓ fm

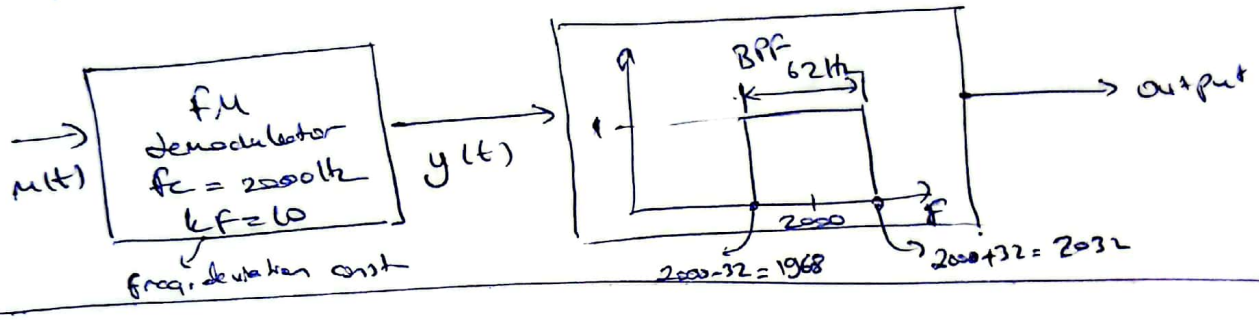
4.18 (FM)

The modulation signal that is the input into an FM modulator is $m(t) = 10\cos(16\pi t)$

The output of the FM modulator is $\theta(t)$

$$y(t) = \underbrace{10}_{A_c}\cos\left[\underbrace{4000\pi t}_{f_c} + 2\pi k_f \int_{-\infty}^{t} m(\tau)\,d\tau\right]$$

where $k_f = 10$ (See figure P-4.18). If the output of the FM modulator is passed through an ideal BPF centered at $f_c = 2000$ with a bandwidth of 62 Hz, determine the power of the frequency components at the output of the filter. What percentage of the transmitter power appears at the output of the BPF?



$$y(t) = A_c \sum_{n=-\infty}^{+\infty} J_n(\beta)\cdot\cos\left(2\pi\left(f_c + n\cdot f_m\right)t\right)$$

$A_c = 10$, $f_c = 2000$, $f_m = 8$, $k_f = 10$

$$\beta = \frac{\Delta f}{f_m} = \frac{k_f \cdot M_p}{f_m} = \frac{k_f \cdot \max|m(t)|}{f_m} = \frac{10\cdot 10}{8} = 12.5$$

$\beta \rightarrow$ B

$$y(t) = 10\cdot\sum_{n=-\infty}^{+\infty} J_n(12.5)\cdot\cos\left(2\pi\left(2000 + n\cdot 8\right)t\right)$$

Freq		freq
J_0 — 2000		J_{-1} — 1992
J_1 J_1 — 2008		J_{-2} — 1984
J_2 J_2 — 2016		J_{-3} — 1976
J_3 J_3 — 2024		J_{-4} — 1968
J_4 J_4 — 2032		

$\uparrow$ n

--4--0--4-

→ power at the output of the BPF.

$$P_y = \frac{10^2}{2}\cdot J_0^2(12.5) + 2\sum_{n=1}^{4} J_n^2(12.5)\cdot\frac{10^2}{2} = 50.02630 = \underline{13.15}$$

β	$J_0(\beta)$	$J_1(\beta)$	$J_2(\beta)$	...
1				
2				
3				
:	Hint			

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$P_{tot} = \frac{A_c^2}{2} = \frac{10^2}{2} = 50.$

↑ total transmitted power

$\frac{P_y}{P_{tot}} = \frac{12.15}{50} = \frac{26.30}{100} \rightarrow \%26.30$

↓

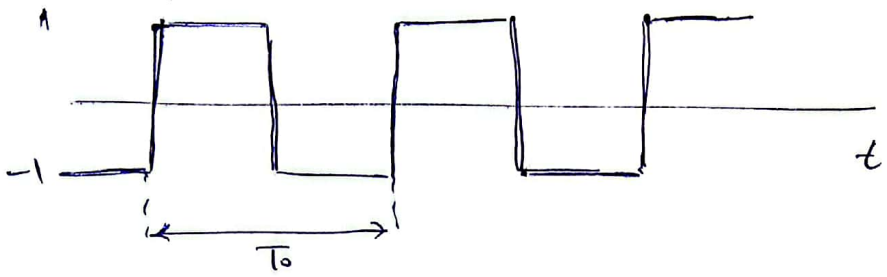
Power at the output of BPF is only %26.30 of transmitted power

- Effective BW → Carson Rule %99

%26.30 → %99 olurdu

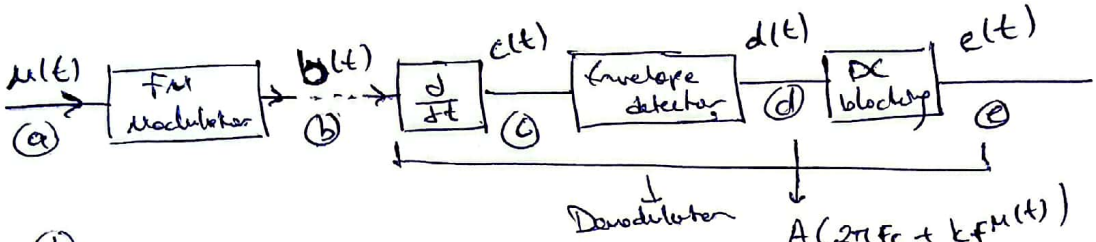
5.4.2

A periodic square wave $m(t)$ (Fig. P.5.4-2a) frequency-modulates a carrier of frequency $f_c = 10\,kHz$ with $\Delta f = 1\,kHz$. The carrier amplitude is A. The resulting FM signal is demodulated, as shown in Fig. P.5.4-2.b. by the method discussed in sec. 5.4 (Fig. 5.11) sketch waveforms. a, b, c, d, e.



(a)

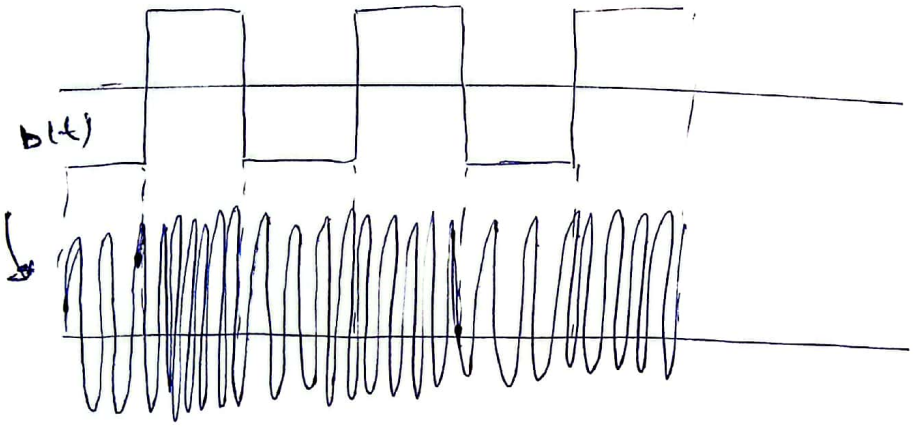
$c(t) = A.\cos(2\pi f_c t)$



(b)

$\max|m(t)| = 1$

$\Delta f = k_f . m_p = k_f = 1k$



(c)

$f_i = f_c + k_f . m(t)$

$f_i = 10k + 1k . m(t)$

$f_{i_{min}} = 10k + 1k\,(m(t)_{min})$

$f_{i_{min}} = 9\,kHz$

$f_{i_{max}} = 10k + 1k\,(m(t)_{max})$

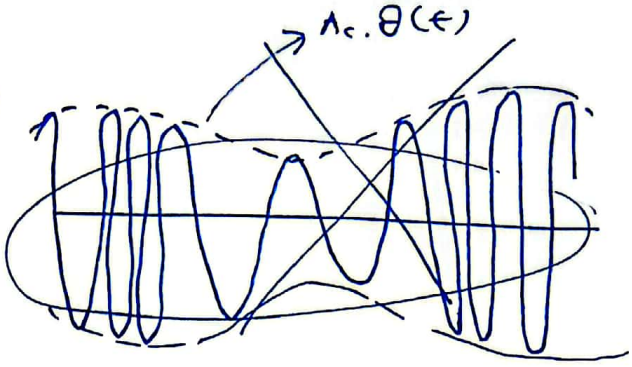
$f_{i_{max}} = 11\,kHz$

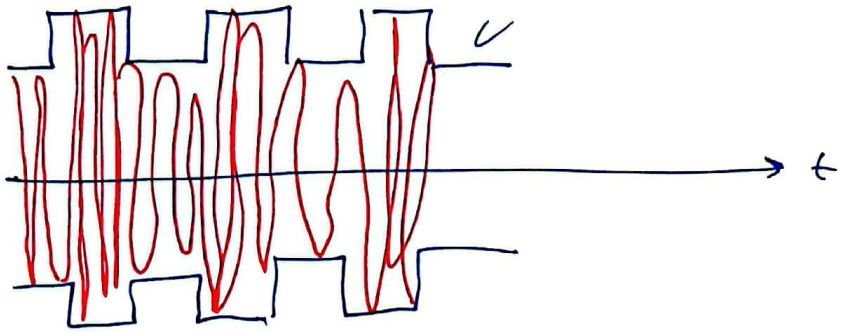
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b(t): $y(t) = A_c \cdot \cos\left(\underbrace{2\pi f_c t + 2\pi k_f \int m(\tau)\, d\tau}_{\theta(t)} \right)$

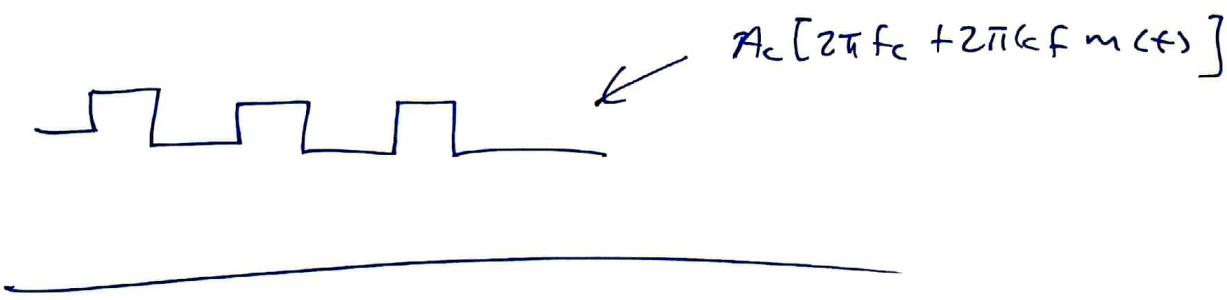
(c) $\frac{d\,y(t)}{dt} = A_c \cdot \theta(t) \sin(\theta(t))$

$A_c [2\pi f_c + 2\pi k_f m(t)] \cdot \sin\theta(t)$

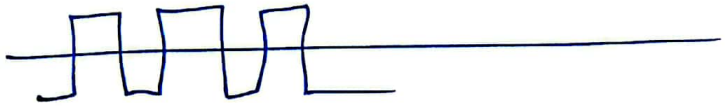




(d)



(e)

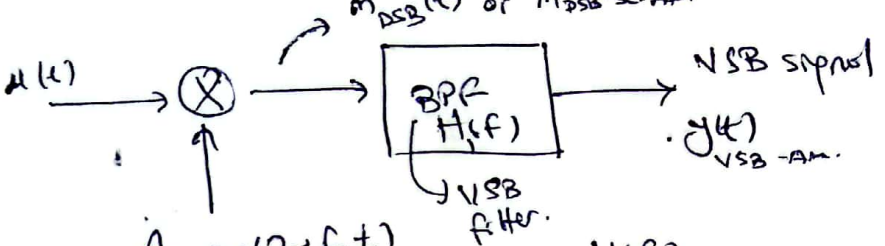


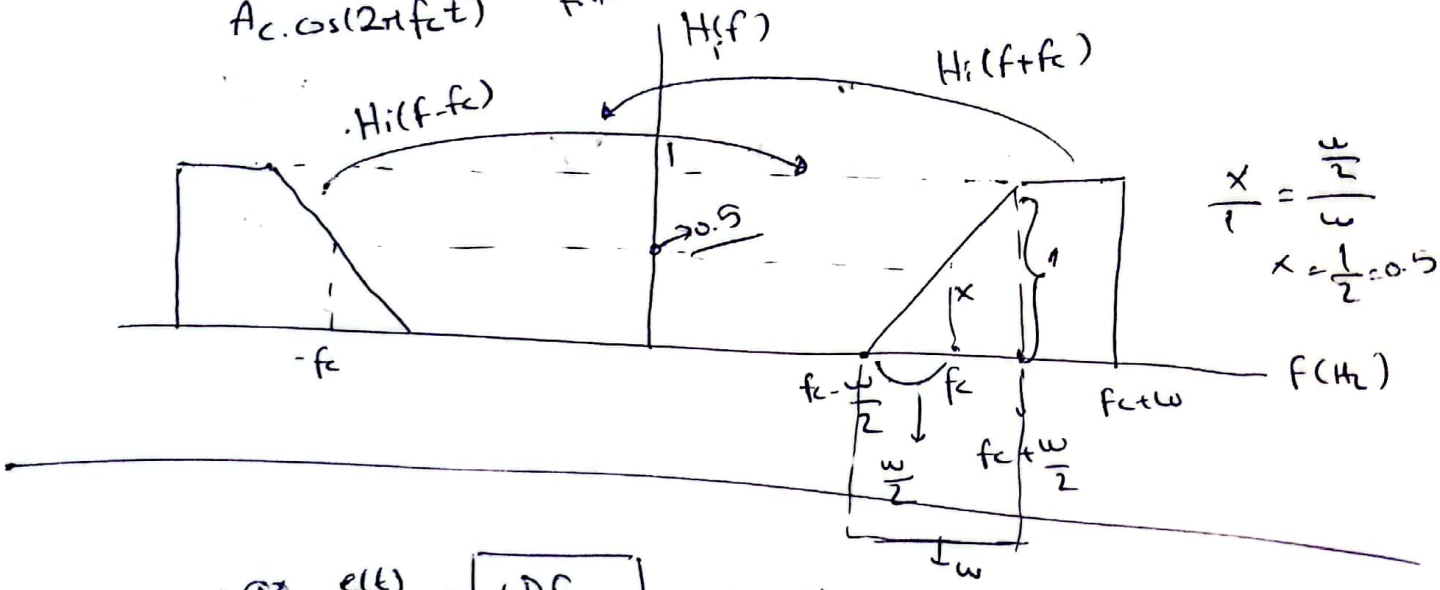
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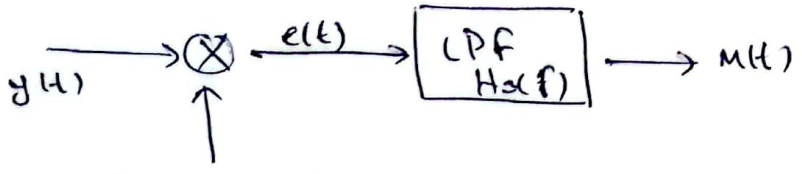
3.21

A vestigial sideband modulation system is shown in figure P.3.21. The bandwidth of the message signal $M(t)$ is W, and the transfer func. of the band pass filter is shown in the figure.

1. Determine $h_l(t)$, which is the low pass equivalent of $h(t)$, where represents the impulse response of the band pass filter.
2. Derive an expression for the modulated signal $y(t)$.







$f \leq B = W$

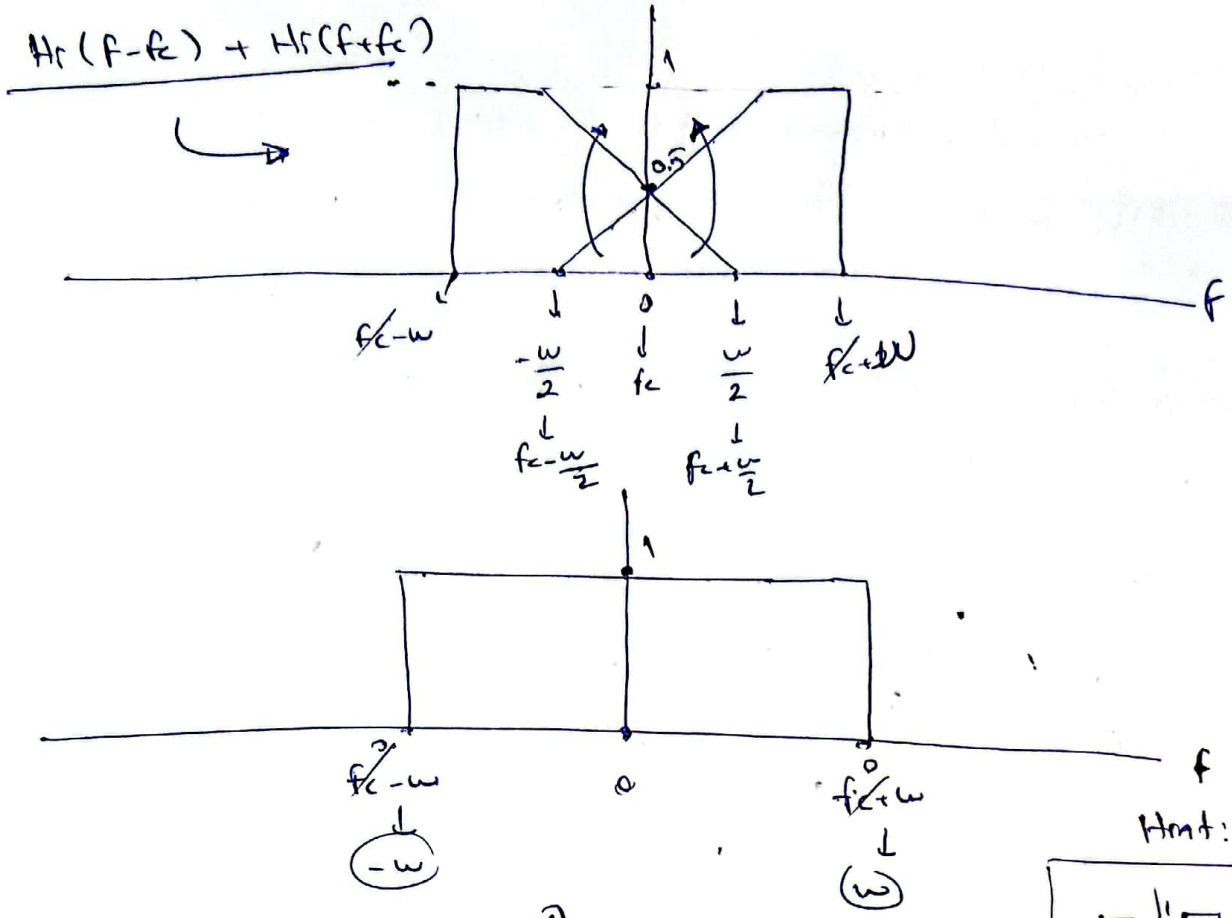
$$H_o(f) = \frac{1}{H_i(f-f_c) + H_i(f+f_c)}$$

$$M(f) = E(f) \cdot H_o(f)$$

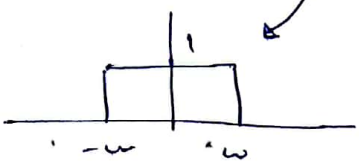
$$M(f) = M(f) \cdot \left(H_i(f-f_c) + H_i(f+f_c)\right) \cdot H_o(f)$$

CamScanner ile tarandı

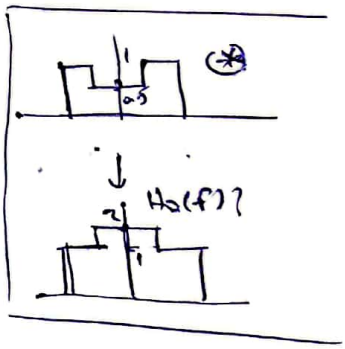
$H_r(f-f_c) + H_r(f+f_c)$



$M(f) = M(f) \cdot (H_r(f-f_c) + H_r(f+f_c)) \cdot H_o(f)$



Hint:



CamScanner ile tarandı